

LYDIA HAILE

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EDUCATION

PhD in Clinical Medicine, Imperial College London 2025-2028

- Department of Infectious Disease Epidemiology London, UK
- Thesis: “Estimating the Epidemiological Impact of Malaria and Polio Vaccine Implementation Through Mathematical Modelling and Statistical Analysis of Surveillance Data”

Master of Public Health (MPH), University of Washington 2019-2022

- Global Health— Health Metrics and Evaluation track Seattle, WA
- Thesis: “Asthma Prevalence, Severity, and Nonfatal Burden by Race/Ethnicity in the United States: A Bayesian Meta-Regression Analysis”

Bachelor of Arts (BA) in Public Health (Honors), Brown University 2015-2019

- GPA: 3.7 Providence, RI
- Honors thesis: “Pediatric Autopsies of Kyiv Region, Ukraine, 2017: Mortality and Discordance between Clinical and Pathological Cause-of-Death data”

EXPERIENCE

Research Assistant and PhD Student, Imperial College London Jul 2025—Present

Department of Infectious Disease Epidemiology London, UK

Vaccine Epidemiology Research Group

- Estimate coverage, immunity, and impact of inactivated polio vaccine on polio immunity and incidence in selected geographies
- Estimate impact of malaria vaccine implementation on cases and deaths in selected geographies

Technical Analyst, Imperial College London Nov 2022—Jul 2025

Department of Infectious Disease Epidemiology London, UK

Malaria Research Group

- Model impact of malaria vaccination on malaria burden for GAVI-funded Vaccine Impact Modelling Consortium
- Utilise a dynamic malaria transmission model, *malariasimulation*, to estimate vaccine impact on malaria cases, deaths and DALYs across Sub-Saharan Africa through 2100
- Provide technical input to GAVI informing optimal allocation of malaria vaccines given limited supply
- Contributed to modelling done for the Strategic Advisory Group of Experts on Immunization (SAGE) and the Immunization and vaccines related implementation research advisory committee (IVIR-AC) to inform malaria vaccine schedule policy guidelines
- Ran models for [Zero Malaria Experience](#), an interactive data installation created in collaboration with Malaria No More UK to advocate for sustained malaria funding

Post-Bachelor Fellow, Institute for Health Metrics and Evaluation Sept 2019—Aug 2022

University of Washington Seattle, WA

- Modelled global burden of hearing loss, musculoskeletal disorders, headaches, and chronic respiratory diseases for the Global Burden of Disease study
- Utilized Bayesian hierarchical models and data adjustment methods to pool fatal and nonfatal data from heterogeneous data sources
- Consulted for WHO Non-Communicable Disease Division to contribute custom analyses of global hearing aid coverage to the World Hearing Report (2021)

Practicum Intern, Ethiopian Public Health Institute Jun 2021—Jan 2022

National Data Management Center Seattle, WA

- Developed national and subnational [health atlas](#) reports on life expectancy, mortality, and leading causes of disease for regional health bureaus
- Collaborated with technical leads to improve the utility of reports for regional health bureaus

- Pulled data to monitor key health metrics for Ethiopian Health Sector Transformation Plan and ran capacity-building training on epidemiologic research methods

Minority Health and International Research Trainee
Brown Global Health Initiative

May 2018—May 2019
Kyiv, Ukraine

- Collected primary data on clinical and pathological tuberculosis symptoms from 635 pediatric autopsy records taken from 4 pediatric hospitals for an improved estimate of tuberculosis burden in the Kyiv region
- Descriptively analysed the relationship between ICD-10 cause of death codes and clinical data for undergraduate thesis

Research Intern
Tikur Anbessa Specialized Hospital

Aug 2018
Addis Ababa, Ethiopia

- Designed a data collection tool for a study of INH Prophylaxis and Tuberculosis prevention in HIV patients and collected several hundred observations from hand-written clinical records

Intern, Institute for Healthcare Improvement
Africa Region

Jun 2017—Aug 2017
Cambridge, MA

- Coordinated the development of a quality improvement course for the Ethiopian Healthcare Quality Initiative, a Ministry of Health project which aimed to reduce maternal mortality by 30% in 5 years

STEER Intern, UC Berkeley School of Public Health
Center for Environmental Research and Children's Health

Jun 2016—Aug 2016
Berkeley, CA

- Conducted literature review for a longitudinal study investigating maternal exposures to pesticides and child health outcomes in the Salinas Valley of California.
- Researched the relationship between maternal pesticide exposure and respiratory and neurodevelopmental health outcomes.

Peer-reviewed Publications

Gaythorpe K, Xiang Li... **Haile LM**, et al. Quantifying relative health impact across Gavi, the Vaccine Alliance's portfolio in 117 countries at the subregional level: a modelling study. *The Lancet*. 2026 May 16; 10.1016/S0140-6736(26)00555-6.

Haile LM, Kamenov K, Briant PS, Orji AU, Steinmetz JD, Abdoli A, et al. Hearing loss prevalence and years lived with disability, 1990–2019: findings from the Global Burden of Disease Study 2019. *The Lancet*. 2021 Mar 13;397(10278):996–1009.

Collaborators G 2019 UHL, **Haile LM**, Orji AU, Reavis KM, Briant PS, Lucas KM, et al. Hearing Loss Prevalence, Years Lived With Disability, and Hearing Aid Use in the United States From 1990 to 2019: Findings From the Global Burden of Disease Study. *Ear and Hearing*. 2024 Feb;45(1):257

Ferreira ML, de Luca K, **Haile LM**, Steinmetz JD, Culbreth GT, Cross M, et al. Global, regional, and national burden of low back pain, 1990–2020, its attributable risk factors, and projections to 2050: a systematic analysis of the Global Burden of Disease Study 2021. *The Lancet Rheumatology*. 2023 Jun 1;5(6):e316–29.

Black RJ, Cross M, **Haile LM**, Culbreth GT, Steinmetz JD, Hagins H, et al. Global, regional, and national burden of rheumatoid arthritis, 1990–2020, and projections to 2050: a systematic analysis of the Global Burden of Disease Study 2021. *The Lancet Rheumatology*. 2023 Oct 1;5(10):e594–610.

Wu AM, Cross M, Elliott JM, Culbreth GT, **Haile LM**, Steinmetz JD, et al. Global, regional, and national burden of neck pain, 1990–2020, and projections to 2050: a systematic analysis of the Global Burden of Disease Study 2021. *The Lancet Rheumatology*. 2024 Mar 1;6(3):e142–55.

Steinmetz JD, Culbreth GT, **Haile LM**, Rafferty Q, Lo J, Fukutaki KG, et al. Global, regional, and national burden of osteoarthritis, 1990–2020 and projections to 2050: a systematic analysis for the Global Burden of Disease Study 2021. *The Lancet Rheumatology*. 2023 Sep 1;5(9):e508–22.

Cross M, Ong KL, Culbreth GT, Steinmetz JD, Cousin E, Lenox H, et al. Global, regional, and national burden of gout, 1990–2020, and projections to 2050: a systematic analysis of the Global Burden of Disease Study 2021. *The Lancet Rheumatology*. 2024 Aug 1;6(8):e507–17.

GBD 2021 Other Musculoskeletal Disorders Collaborators. Global, regional, and national burden of other musculoskeletal disorders, 1990–2020, and projections to 2050: a systematic analysis of the Global Burden of Disease Study 2021. *Lancet Rheumatol*. 2023 Nov;5(11):e670–82.

Murray CJL, Abbafati C, Abbas KM, Abbasi M, Abbasi-Kangevari M, Abd-Allah F, et al. Five insights from the Global Burden of Disease Study 2019. *The Lancet*. 2020 Oct 17;396(10258):1135–59.

Vos T, Lim SS, Abbafati C, Abbas KM, Abbasi M, Abbasifard M, et al. Global burden of 369 diseases and injuries in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019. *The Lancet*. 2020 Oct 17;396(10258):1204–22.

Submitted and Pre-Print Publications

Haile L, McCain K, Schmit N, Topazian H, Rouamba T, Winskill P, Ghani A, Okell L. Long-term Impact of Vaccine Introduction on Malaria Cases and Deaths in Sub-Saharan Africa: A Modelling Study. *Lancet Global Health* (under review).

Additional Publications

Winskill P, **Haile L**, Ruybal-Pesántez S, Simmons O, Topazian H, Okell L. Rapid response modelled estimates of the effect of the US global aid freeze on President's Malaria Initiative impact in sub-Saharan Africa. 2025 Mar 5; Available from: <https://hdl.handle.net/10044/1/118190>

Hearing Loss, Social Isolation, and the COVID-19 Pandemic. Think Global Health [Internet]; Available from: <https://www.thinkglobalhealth.org/article/hearing-loss-social-isolation-and-covid-19-pandemic>

Mapping the financial and disease burden of hearing loss and associated interventions. ENT & Audiology News; Available from: <https://www.entandaudiologynews.com/features/audiology-features/post/mapping-the-financial-and-disease-burden-of-hearing-loss-and-associated-interventions>

Obstacles to Healing Asthma. Think Global Health [Internet]; Available from: <https://www.thinkglobalhealth.org/article/obstacles-healing-asthma>

National Health Atlas – National Data Management Center for Health. Available from: <https://ndmc.ephi.gov.et/national-health-atlas/>

World Health Organization. 2021 World report on hearing. Available from: <https://www.who.int/publications/i/item/9789240020481>

Oral and Poster Presentations

Presentation: **Lydia Haile**, Isobel Blake. Updates on polio modelling work from Imperial. World Health Organization Polio Research Committee meeting. September 2026.

Presentation: **Lydia Haile**, Lucy Okell, Nora Schmit, Pete Winskill. Public health impact and cost-effectiveness of 3-dose versus 4-dose malaria vaccine schedule. London School of Hygiene and Tropical Medicine Malaria Centre Away Day. July 2026.

Presentation: **Lydia Haile**, Isobel Blake. Estimating subnational coverage of the inactivated polio vaccine in routine immunisation and population immunity against serotype-2 poliomyelitis in Nigeria, 2021–2025. South African Centre for Epidemiological Modelling and Analysis modelling fellows. July 2026.

Poster: **Lydia Haile**, Isobel Blake. Subnational IPV coverage in Nigeria, 2021–2024: statistical analysis of routine immunization surveillance data. Empower Imperial: Connecting Black Researchers. June 2026.

Presentation: **Lydia Haile**, Isobel Blake. Inactivated poliovirus vaccine coverage and population immunity in Nigeria, 2021-2024: statistical analysis of acute flaccid paralysis data. Imperial School of Public Health PhD Symposium. June 2026.

Presentation: **Lydia Haile**. Modelling the impact of vaccine implementation. Jimma University Infectious Disease Modelling Workshop (virtual). May 2026.

Presentation: **Lydia Haile**. Reflections on a career in health data science. Ethiopian Data Analytics and Modelling Consortium Women in Health Data Day. April 2026.

Presentation: **Lydia Haile**. Introduction to infectious disease modelling. Ethiopian Data Analytics and Modelling Consortium Workshop. April 2026.

Presentation: **Lydia Haile**, Kelly McCain, Hillary Topazian, Nora Schmit, Toussaint Rouamba, Katy Gaythorpe, Azra Ghani, Pete Winskill, Lucy Okell. Potential Impact of Vaccine Introduction on Malaria Cases and Deaths in Sub-Saharan Africa: A Modelling Study. American Society for Tropical Medicine and Hygiene. November 2025.

Presentation: **Lydia Haile**, Andrew Shattock, Romain Glele, Melissa Penny, Lucy Okell. Modelling malaria vaccine impact. Vaccine Impact Modelling Consortium Annual Meeting. June 2025.

Presentation: **Lydia Haile**, Nora Schmit. Preliminary analyses of public health impact and cost effectiveness [of malaria vaccines]. WHO Immunization and vaccines related implementation research advisory committee (IVIR-AC). July 2025.

Poster: **Lydia Haile**, Nora Schmit, Kelly McCain, Hillary M. Topazian, Giovanni D. Charles, Katherina Hauck, Azra Ghani, Peter Winskill, Lucy Okell. Modelling the Impact of RTS, and R21 vaccination on endemic settings in Sub-Saharan Africa. 8th Annual Multilateral Initiative on Malaria (MIM) Society Conference. April 2024.

Presentation: **Lydia Haile**, Lucy Okell. Modelling malaria vaccine impact. Vaccine Impact Modelling Consortium Annual Meeting. September 2023.

Presentation: **Lydia Haile**. Population Hearing Healthcare Annual Meeting, 2021. Updates on hearing loss prevalence estimation for the Global Burden of Disease study. March 2021.

Presentation: **Haile L**, Orji A, Briant P, Adelson J, Davis A, Vos T. Gerontological Sciences of America Annual Meeting. Updates on Hearing From the Global Burden of Disease Study. November 2020.

Presentation: **Lydia Haile**. Population Hearing Healthcare Annual Meeting, 2020. Estimation of hearing loss prevalence for the Global Burden of Disease study. March 2020.

Public Engagement

Honoring Deaf History Month: a conversation with researcher Lydia Haile about IHME's hearing loss study. [IHME](#).

David Beckham joins scientists in video launched at Zero Malaria Experience. [David Beckham joins scientists in video launched at Zero Malaria Experience | Imperial News | Imperial College London](#)

MRC Centre for Global Infectious Disease Analysis Science in Context.
<https://www.linkedin.com/feed/update/urn:li:activity:7260310708981055491/>

Academic and Professional Service

IHME DEI Council Research Committee Lead

2021-2022

- Led the development of equity in research guidelines for IHME, a staff resource used to incorporate an equity perspective into data collection, analysis, and external communication
- Contributed to equity and inclusion programming throughout the institute

IHME Black Employee Resource Group Leader

2020-2022

- Led informal group for Black staff and implemented cultural programming

Member of Ye Tena Weg Ethiopian Health Professional Association

Teaching Experience

Ethiopian Data Analytics and Modelling Consortium IDM Workshop Demonstrator	April and Sept 2026
Short Course on Mathematical Modelling of Infectious Diseases Demonstrator, Imperial College London	2025
MSc in Epidemiology Course Demonstrator, Imperial College London	2024
Malaria modelling training demonstrator, Imperial College London	2024
Teaching Assistant, Burden of Disease in Developing Countries, Brown University	2017-2019

Awards and Nominations

Awarded: Imperial College President's Medal for excellence in research impact (group)	2026
Nominated: Ear and Hearing Editor's Award	2023
Awarded: Miliken Institute Future Health Leader, Brown University, Miliken Institute	2018
Awarded: University of Washington Graduate School Outstanding Master's Thesis Award	2022
Awarded: Institute for Health Metrics and Evaluation Community Builder Award	2022
Nominated: Institute for Health Metrics and Evaluation Innovator of the Year Award	2022
Nominated: Institute for Health Metrics and Evaluation Community Builder Award	2021

Technical skills

Languages: Amharic (Conversational), Spanish (Conversational), English (fluent)
Technology: R (Advanced), Git (Advanced), Stata (Proficient), Python (Beginner), Redcap (Proficient)